

Topic: A+ - Troubleshooting and Helpdesk

ASSIGNMENT 1

Task (1):

1. **Computer won't power on:** A multimeter checks the power supply's output voltages to confirm whether the PSU is delivering the correct power or has failed.
2. **Smartphone won't charge:** A multimeter measures the charger, USB cable, or battery voltage to identify whether the charging accessories or battery are faulty.

3. **Laptop or smartphone has a suspected broken connection:** A multimeter's continuity mode tests cables, connectors, or fuses to detect open circuits or damaged connections.

Task (2):

POST Card Demo Summary:

- In the demo, the POST card displays changing hexadecimal codes as the BIOS checks different motherboard components.
- A POST card shows two-digit codes while the computer starts, helping check the motherboard hardware.
- The codes indicate which part of the POST (Power-On Self-Test) the computer has reached.

Task (3):

HWMonitor Software:

Sensor	Value	Min	Max
Dell Inc. 0PRMWD			
Fans			
Fan 0	0 RPM	0 RPM	0 RPM
Utilization			
Physical Memory Load	65.0 %	63.0 %	66.0 %
Available Physical Me...	2.7 GB	2.7 GB	2.9 GB
Virtual Memory Load	53.5 %	51.8 %	54.1 %
Available Virtual Mem...	4.7 GB	4.6 GB	4.9 GB
Intel Core i5 10210U			
Voltages			
VID (Max)	0.669 V	0.649 V	1.173 V
Temperatures			
Cores (Max)	52.0 °C	51.0 °C	83.0 °C
Package	52.0 °C	51.0 °C	86.0 °C
Powers			
Package	1.74 W	1.53 W	21.25 W
IA Cores	0.51 W	0.31 W	18.42 W
GT	0.00 W	0.00 W	0.00 W
DRAM	0.39 W	0.36 W	0.86 W
Power Max (PL1)	28.00 W	28.00 W	28.00 W
Short Power Max (PL2)	51.00 W	51.00 W	51.00 W
Max Peak Power (PL4)	90.00 W	90.00 W	90.00 W
Currents			
IccMax	85.00 A	85.00 A	85.00 A
Clocks			
Cores (Max)	1097.8 MHz	795.9 MHz	3986.7 MHz
CPU BCLK	99.8 MHz	96.0 MHz	100.4 MHz
LLC/Ring	1297.4 MHz	497.2 MHz	3792.1 MHz
Memory	667.8 MHz	664.0 MHz	670.5 MHz
Utilization			
Processor	3.7 %	2.2 %	33.2 %
TOSHIBA MQMABF100 (Y90FTF95T)			
Temperatures			
Assembly	32.0 °C	32.0 °C	32.0 °C
Counters			
Power-On-Hours	1981 hrs	1981 hrs	1981 hrs
Power Cycle Count	2552	2552	2552
Utilization			
Used Space (e)	10.8 %	10.8 %	10.8 %
Available Capacity (e)	134.1 GB	134.1 GB	134.1 GB
Used Space (f)	20.2 %	20.2 %	20.2 %
Available Capacity (f)	207.7 GB	207.7 GB	207.7 GB
Used Space (g)	24.1 %	24.1 %	24.1 %
Available Capacity (g)	197.7 GB	197.7 GB	197.7 GB
Used Space (h)	80.7 %	80.7 %	80.7 %
Available Capacity (h)	50.3 GB	50.3 GB	50.3 GB
Activity	0.1 %	0.0 %	1.2 %
Speed			
Read Rate	0.00 MB/s	0.00 MB/s	0.00 MB/s
Write Rate	0.00 MB/s	0.00 MB/s	0.00 MB/s
ZEUS-m.2nvme0220705TNSR128G			
Temperatures			
Assembly	45.0 °C	45.0 °C	45.0 °C
Counters			
Power-On-Hours	720 hrs	720 hrs	720 hrs
Power Cycle Count	1016	1016	1016
Health Status	98 %	98 %	98 %
Utilization			
Used Space (c)	86.3 %	86.3 %	86.3 %
Available Capacity (c)	16.2 GB	16.2 GB	16.2 GB
Activity	3.5 %	0.0 %	10.4 %
Speed			
Read Rate	0.42 MB/s	0.00 MB/s	2041 MB/s
Write Rate	2.81 MB/s	0.00 MB/s	2.81 MB/s
AMD Radeon 610 Series			
Voltages			
GPU	0.850 V	0.850 V	0.875 V
Temperatures			
GPU	49.0 °C	49.0 °C	58.0 °C
Clocks			
Graphics	300.0 MHz	300.0 MHz	400.0 MHz
Memory	150.0 MHz	150.0 MHz	1000.0 MHz
Utilization			
GPU	0.0 %	0.0 %	29.0 %
Video Engine	0.0 %	0.0 %	0.0 %
3D	0.0 %	0.0 %	3.1 %
Copy	0.0 %	0.0 %	2.4 %
Compute	0.0 %	0.0 %	0.0 %
Dedicated Memory	98.0 MB	69.1 MB	122.7 MB
Shared Memory	25.4 MB	8.2 MB	36.9 MB
Memory	4.8 %	3.4 %	6.0 %
Speed			
PCIe Link Speed	2.50 GT/s	2.50 GT/s	8.00 GT/s
Microsoft Basic Display Adapter			
Temperatures			
GPU	55.0 °C	51.0 °C	62.0 °C
Clocks			
Graphics	0.0 MHz	0.0 MHz	0.0 MHz
Utilization			
GPU	0.0 %	0.0 %	0.0 %

Task (4):

Tool	What it checks	Example Problem Detected
Multimeter	Checks voltage, current, resistance, and electrical connections.	Finds a faulty power supply or broken cable.
POST Card	Checks the motherboard during startup and displays error codes.	Detects a RAM or CPU hardware problem during boot.
Software Diagnostics	Checks temperatures, storage health, memory, and hardware performance.	Finds an overheating CPU or unhealthy hard drive/SSD.
